# Lctr 22: Sampling

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Focus: Chapter 7 (Pages 514–555)

## 1 The Big Picture: From Continuous Time to Discrete Time

**Why This Matters**

Nearly every modern signal-processing system converts continuous-time signals into discrete-time signals at some point. Your phone digitizes your voice before compressing and transmitting it. A digital camera samples a spatially continuous scene onto a finite grid of pixels. An MRI scanner collects discrete frequency-domain measurements of a continuous spatial distribution.

The central question of this lecture is: **Under what conditions can we recover a continuous-time signal perfectly from a set of equally spaced samples?** The answer is the *sampling theorem*, one of the most important results in all of signals and systems.

**Roadmap for this lecture:**

1. Introduce impulse-train sampling and derive its frequency-domain effect.
2. State and interpret the sampling theorem.
3. Show how to reconstruct a signal from its samples (band-limited interpolation).
4. Explore what happens when we violate the sampling theorem (aliasing).
5. Connect sampling to discrete-time processing of continuous-time signals.

## 2 Impulse-Train Sampling

### 2.1 The Sampling Model

To study sampling mathematically, we model the process as multiplying $x(t)$ by a periodic impulse train $p(t)$ with period $T$:

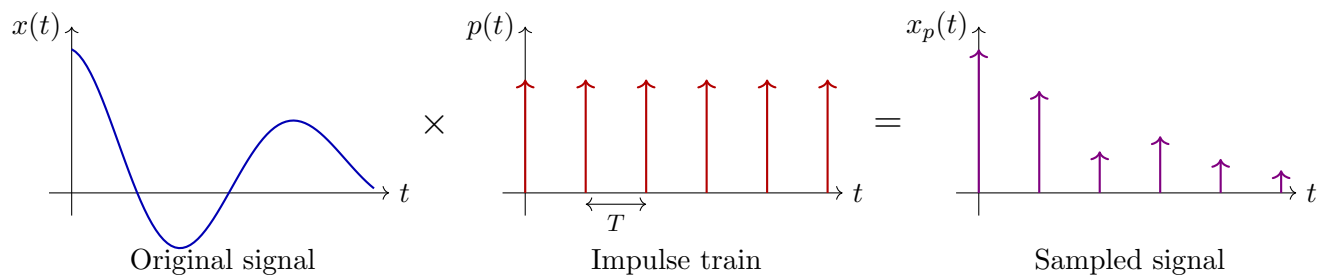
$$x_p(t) \;=\; x(t)\,p(t), \qquad p(t) \;=\; \sum_{n=-\infty}^{+\infty} \delta(t - nT). \tag{1}$$

The period $T$ is the **sampling period** (time between samples), and the fundamental frequency $\omega_s = 2\pi/T$ is the **sampling frequency** (how many radians per second we sample at).

Because multiplying by an impulse simply captures the signal value at that instant ($x(t)\,\delta(t - nT) = x(nT)\,\delta(t - nT)$), the sampled signal is an impulse train whose amplitudes are the sample values:

$$x_p(t) \;=\; \sum_{n=-\infty}^{+\infty} x(nT)\,\delta(t - nT). \tag{2}$$

Think of it this way: we "grab" the value of $x(t)$ at every multiple of $T$ and store it as the height of an impulse at that location.




## 2.2 Frequency-Domain Effect of Sampling

From the multiplication property of the Fourier transform, multiplying in time corresponds to convolution in frequency (scaled by $1/2\pi$). The Fourier transform of the impulse train is itself an impulse train:

$$P(j\omega) \;=\; \frac{2\pi}{T} \sum_{k=-\infty}^{+\infty} \delta(\omega - k\omega_s). \tag{3}$$

Convolving $X(j\omega)$ with this impulse train produces shifted replicas:

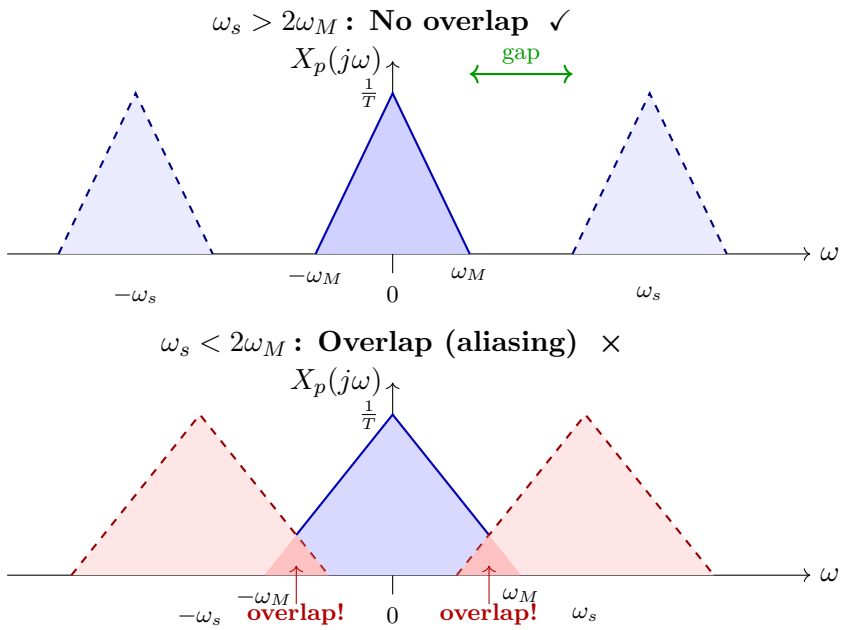
$$\boxed{X_p(j\omega) \;=\; \frac{1}{T} \sum_{k=-\infty}^{+\infty} X\big(j(\omega - k\omega_s)\big).} \tag{4}$$

In plain English: the spectrum of the sampled signal is just the original spectrum $X(j\omega)$ *copied and pasted* at every integer multiple of $\omega_s$, and each copy is scaled by $1/T$.

**Key Insight**

Sampling in time creates **periodic copies (replicas)** of the original spectrum $X(j\omega)$, shifted to every integer multiple of $\omega_s$ and scaled by $1/T$. The entire content of the sampling theorem comes down to one question: **do these copies overlap, or not?**

Let us visualize both cases. Suppose $X(j\omega)$ is a simple triangle living between $-\omega_M$ and $+\omega_M$.




## 3 The Sampling Theorem

**The Sampling Theorem (Shannon / Nyquist)**

Let $x(t)$ be a **band-limited** signal with

$$X(j\omega) = 0 \quad \text{for} \quad |\omega| > \omega_M.$$

Then $x(t)$ is **uniquely determined** by its samples $x(nT)$, $n = 0, \pm 1, \pm 2, \ldots$, if

$$\boxed{\omega_s > 2\omega_M} \qquad \text{where} \quad \omega_s = \frac{2\pi}{T}.$$

Given these samples, $x(t)$ can be reconstructed exactly by passing an impulse train (whose impulse amplitudes equal the sample values) through an ideal lowpass filter with gain $T$ and cutoff frequency $\omega_c$ satisfying $\omega_M < \omega_c < \omega_s - \omega_M$.

The frequency $2\omega_M$ is called the **Nyquist rate**: the minimum sampling frequency you need. The frequency $\omega_M$ (half the Nyquist rate) is often called the **Nyquist frequency**.

**Key Insight**

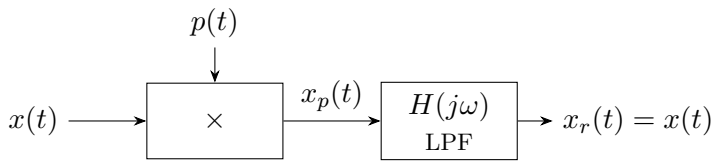
The sampling theorem says two things:

1. **Uniqueness:** If you sample fast enough, no information is lost.
2. **Reconstruction:** You can get $x(t)$ back exactly using an ideal lowpass filter.

The condition $\omega_s > 2\omega_M$ is a *strict* inequality. Sampling at exactly twice the highest frequency is *not* sufficient in general (Example 22.3 shows why).

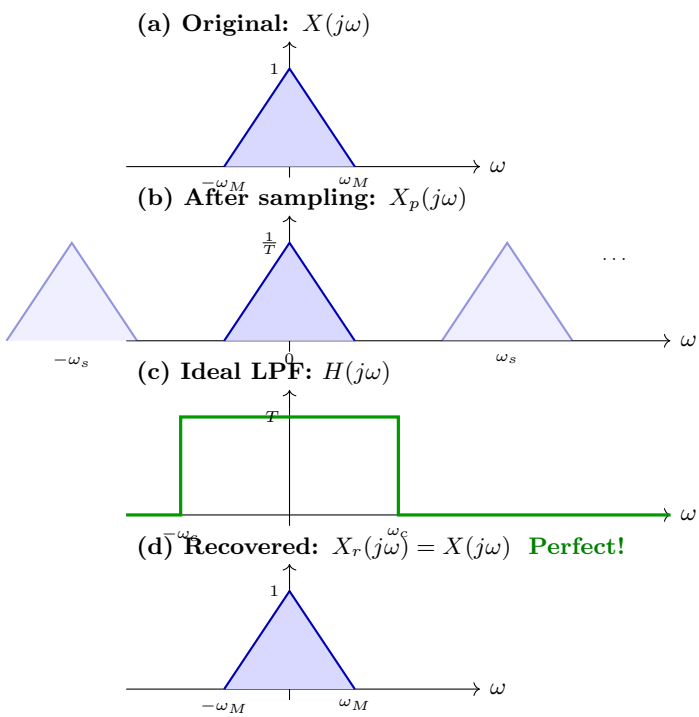
### 3.1 The Reconstruction System

The ideal lowpass filter $H(j\omega)$ isolates the center copy of the spectrum and throws away all the other replicas.




With cutoff at $\omega_c = \omega_s/2$: $H(j\omega) = T$ for $|\omega| < \omega_s/2$, and $H(j\omega) = 0$ for $|\omega| > \omega_s/2$.

Here is the full picture of what happens at each stage in the frequency domain:




## 4 Worked Examples

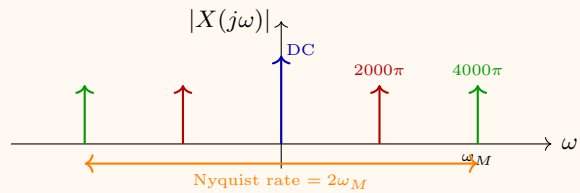
**Example 22.1(a): Nyquist Rate of a Sum of Sinusoids**

**Problem.** Find the Nyquist rate of $x(t) = 1 + \cos(2000\pi t) + \sin(4000\pi t)$.

The three frequency components are at 0, $2000\pi$, and $4000\pi$ rad/s. The highest is $\omega_M = 4000\pi$ rad/s $= 2000$ Hz.

$$\text{Nyquist rate} = 2\omega_M = 8000\pi \text{ rad/s} = 4000 \text{ Hz}.$$

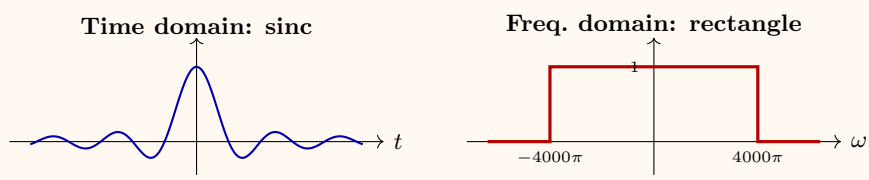
We need to sample faster than 4000 times per second.




## Example 22.1(b): Nyquist Rate of a Sinc Function

**Problem.** Find the Nyquist rate of $x(t) = \dfrac{\sin(4000\pi t)}{\pi t}$.

We rewrite: $x(t) = 4000 \cdot \frac{\sin(4000\pi t)}{4000\pi t} = 4000\ \mathrm{sinc}(4000\,t)$, where $\mathrm{sinc}(u) \triangleq \sin(\pi u)/(\pi u)$.
Its Fourier transform is a **rectangle**: $X(j\omega) = 1$ for $|\omega| < 4000\pi$, and 0 otherwise. So $\omega_M = 4000\pi$ and the Nyquist rate is $8000\pi$ rad/s $= 4000$ Hz.



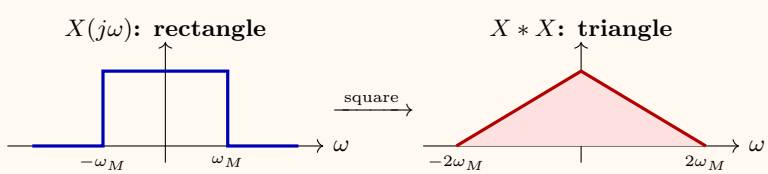

## Example 22.1(c): Nyquist Rate of a Squared Sinc

**Problem.** Find the Nyquist rate of $x(t) = \left(\dfrac{\sin(4000\pi t)}{\pi t}\right)^2$.

First, recognize what is being squared. From part (b) we know that $\frac{\sin(4000\pi t)}{\pi t} = 4000\ \mathrm{sinc}(4000\,t)$ is a sinc function with bandwidth $\omega_M = 4000\pi$.
Now we are squaring that entire sinc function. Squaring in the time domain corresponds to convolving the spectrum with itself in the frequency domain (since $x^2(t) \leftrightarrow \frac{1}{2\pi} X(j\omega) * X(j\omega)$). The convolution of two rectangles, each of width $2 \times 4000\pi = 8000\pi$, produces a **triangle** of total width $2 \times 8000\pi = 16000\pi$. Therefore $\omega_M = 8000\pi$ and:

$$\text{Nyquist rate} = 2\omega_M = 16000\pi \text{ rad/s} = 8000 \text{ Hz}.$$

The lesson: squaring a signal **doubles** the bandwidth!




## Example 22.2: Checking Whether a Sampling Period Works

**Problem.** A signal $x(t)$ is the output of an ideal lowpass filter with cutoff frequency $\omega_c = 1000\pi$ rad/s. Which of the following sampling periods guarantee that $x(t)$ can be perfectly

reconstructed from its samples?

(a) $T = 0.5 \times 10^{-3}\,\text{s}$

(b) $T = 2 \times 10^{-3}\,\text{s}$

(c) $T = 10^{-4}\,\text{s}$

**Step 1: Identify the highest frequency.**
Since $x(t)$ comes out of an ideal lowpass filter with cutoff $\omega_c = 1000\pi$ rad/s, we know $X(j\omega) = 0$ for $|\omega| > 1000\pi$. So the highest frequency present in $x(t)$ is:

$$\omega_M = 1000\pi \text{ rad/s} \qquad \big(\text{equivalently, } f_M = 500 \text{ Hz}\big).$$

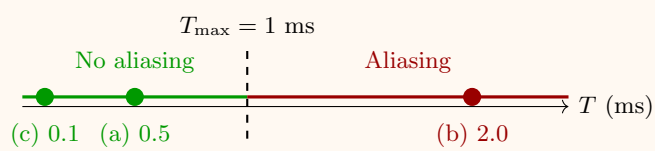
**Step 2: Compute the Nyquist rate.**
The sampling theorem requires $\omega_s > 2\omega_M$:

$$\omega_s > 2 \times 1000\pi = 2000\pi \text{ rad/s} \qquad \big(\text{equivalently, } f_s > 1000 \text{ Hz}\big).$$

**Step 3: Convert to a constraint on $T$.**
Since $\omega_s = 2\pi/T$, the condition $\omega_s > 2000\pi$ becomes:

$$\frac{2\pi}{T} > 2000\pi \quad \Longrightarrow \quad T < \frac{2\pi}{2000\pi} = \frac{1}{1000} = 10^{-3}\text{ s} = 1\text{ ms}.$$

Any sampling period *less than* 1 ms will work.
**Step 4: Check each option.**

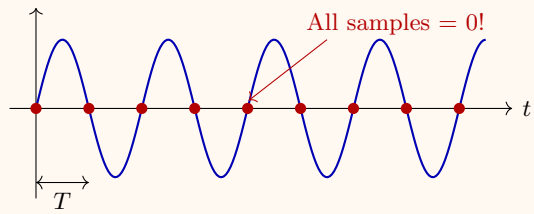



- **(a)** $T = 0.5$ ms $< 1$ ms: $\omega_s = 2\pi/(0.5 \times 10^{-3}) = 4000\pi > 2000\pi$. ✓ This works. We are sampling at twice the Nyquist rate (oversampled by $2\times$).
- **(b)** $T = 2$ ms $> 1$ ms: $\omega_s = 2\pi/(2 \times 10^{-3}) = 1000\pi < 2000\pi$. × This is too slow. The sampling frequency is only half of what we need, so aliasing will distort the signal.
- **(c)** $T = 0.1$ ms $< 1$ ms: $\omega_s = 2\pi/(10^{-4}) = 20000\pi \gg 2000\pi$. ✓ This works easily. We are sampling at $10\times$ the Nyquist rate.

**Example 22.3: Sampling at Exactly the Nyquist Rate Fails**

**Problem.** Consider $x(t) = \sin\big(\frac{\omega_s}{2}t\big)$ sampled at frequency $\omega_s$ (exactly two samples per cycle).

**Solution.** The samples are $x(nT) = \sin\big(\frac{\omega_s}{2} \cdot \frac{2\pi n}{\omega_s}\big) = \sin(\pi n) = 0$ for every integer $n$. Every sample is zero! The sampler sees nothing.




This is why $\omega_s$ must be *strictly greater than* $2\omega_M$.

## 5 Reconstruction via Band-Limited Interpolation

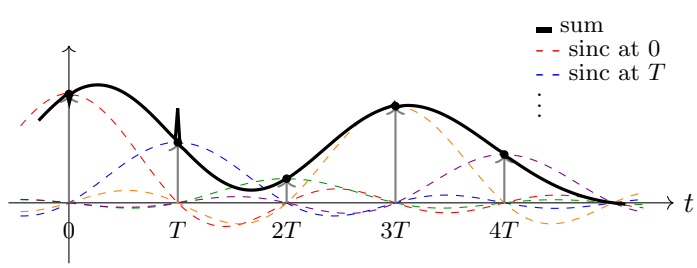
In the time domain, the output of the ideal lowpass reconstruction filter is:

$$x_r(t) = \sum_{n=-\infty}^{+\infty} x(nT)\, h(t - nT), \qquad h(t) = \frac{\sin(\pi t/T)}{\pi t/T}. \tag{5}$$

So the reconstruction formula is:

$$\boxed{x_r(t) = \sum_{n=-\infty}^{+\infty} x(nT)\, \frac{\sin\big[\pi(t - nT)/T\big]}{\pi(t - nT)/T}.} \tag{6}$$

Place a sinc at every sample point, scale it by the sample value, and add them all up. Each sinc equals 1 at its own sample time and 0 at every other sample time.



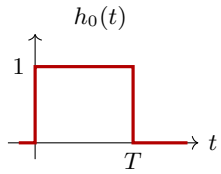

**Key Insight**

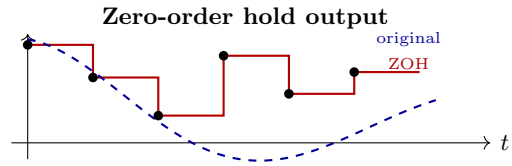
Equation (6) is the **ideal band-limited interpolation formula**. In practice, the sinc extends to $\pm\infty$ (you would need infinitely many samples), so real systems use approximate methods: zero-order holds, linear interpolation, or windowed-sinc filters.

## 6 Practical Interpolation: Zero-Order and First-Order Holds

### 6.1 Zero-Order Hold (Staircase)

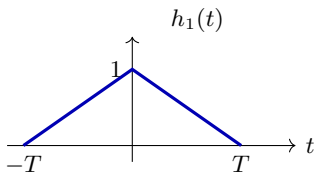
Hold each sample value constant until the next sample arrives. Uses a rectangular pulse $h_0(t)$ of width $T$:

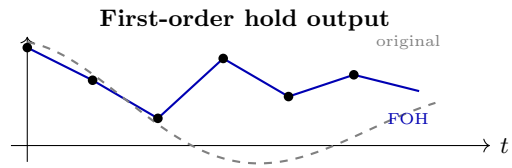



### 6.2 First-Order Hold (Connect-the-Dots)

Connect adjacent samples with straight lines. Uses a triangular pulse $h_1(t)$:




Neither method is exact. Both approximate the ideal sinc interpolation.

## 7 The Effect of Undersampling: Aliasing

When $\omega_s < 2\omega_M$, the shifted replicas of $X(j\omega)$ overlap, and the original spectrum cannot be separated out by lowpass filtering. This distortion is called **aliasing**: high-frequency components masquerade as lower frequencies.

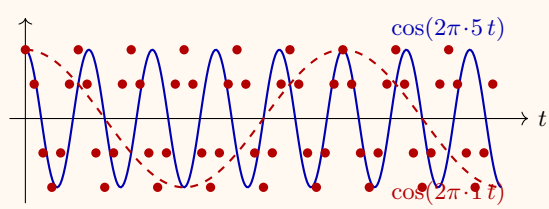
**Warning**

Aliasing is **irreversible**. Once spectral replicas overlap, there is no way to undo the damage. The only way to prevent aliasing is to ensure $\omega_s > 2\omega_M$ *before* sampling, or to pre-filter the signal to remove frequencies above $\omega_s/2$ (an **anti-aliasing filter**).

### 7.1 Aliasing of a Sinusoid

**Example 22.4: A 5 Hz Cosine Sampled at 6 Hz**

Consider $x(t) = \cos(2\pi \cdot 5\,t)$ sampled at $f_s = 6\,\text{Hz}$. The Nyquist rate is $2f_0 = 10\,\text{Hz}$. Since $f_s = 6 < 10$, we are undersampling. The aliased frequency is $f_{\text{alias}} = f_s - f_0 = 6 - 5 = 1\,\text{Hz}$.

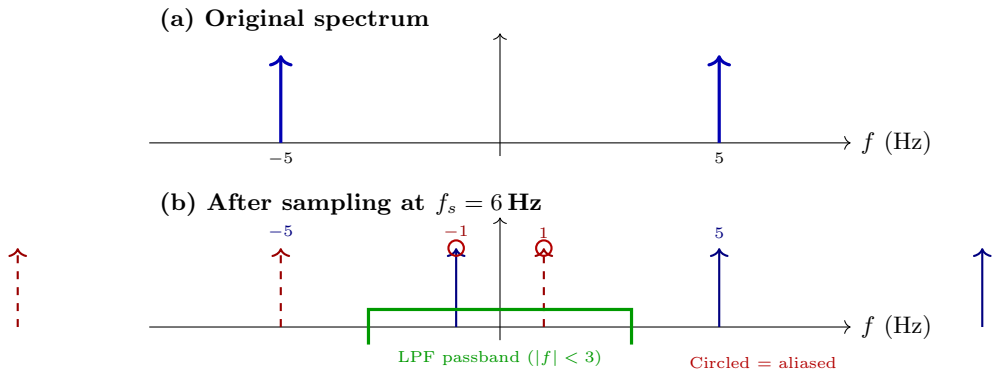



Both curves pass through the same sample points!

The 5 Hz and 1 Hz signals produce the same samples. The lowpass filter picks the lower one, so the 5 Hz tone is aliased to 1 Hz.

### 7.2 Aliasing in the Frequency Domain

The original spectrum has impulses at $\pm 5\,\text{Hz}$. After sampling at $6\,\text{Hz}$, replicas appear at $\pm 5 + 6k$. The replicas at $\pm 1\,\text{Hz}$ land inside the lowpass filter:



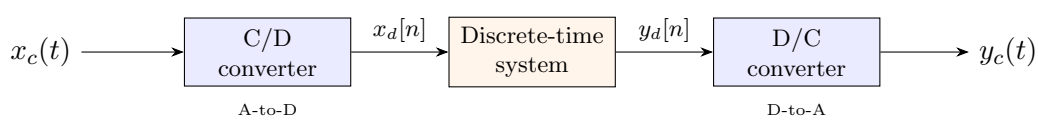

### 7.3 Everyday Examples of Aliasing

**Wagon-wheel effect in movies:** A movie camera captures frames at 24 fps. If a wagon wheel spins near 12 revolutions per second, it can appear to stand still or rotate backward. The rotation frequency exceeds half the frame rate.

**Moiré patterns:** Photographing a fine-striped shirt with a coarse pixel grid produces wavy interference patterns: spatial aliasing.

## 8 Discrete-Time Processing of Continuous-Time Signals

One of the most powerful applications of sampling is using discrete-time systems (computers, DSP chips) to process continuous-time signals:

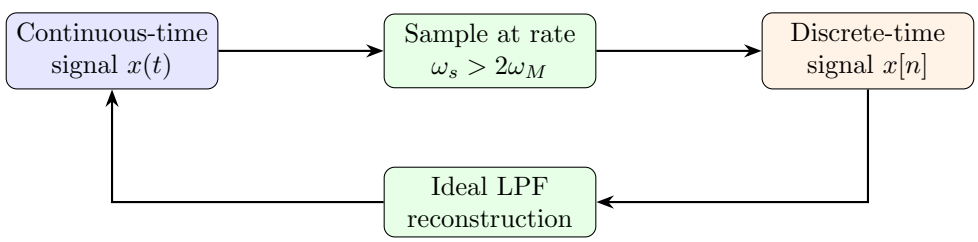



The C/D converter creates $x_d[n] = x_c(nT)$. The D/C converter reconstructs $y_c(t)$ from $y_d[n]$ via lowpass filtering. If $x_c(t)$ is band-limited and the sampling theorem is satisfied, this entire cascade behaves like a continuous-time LTI system. The discrete-time frequency $\Omega$ and the continuous-time frequency $\omega$ are related by $\Omega = \omega T$, so the full range $|\omega| < \omega_s/2$ maps onto $|\Omega| < \pi$.

# 9 Summary of Key Formulas

| **Concept** | **Formula / Condition** |
|---|---|
| Sampling period / frequency | $T = 2\pi/\omega_s$ |
| Nyquist rate | $2\omega_M$ (must be exceeded by $\omega_s$) |
| Sampling condition | $\omega_s > 2\omega_M$ (strict inequality!) |
| Spectrum of sampled signal | $X_p(j\omega) = \frac{1}{T}\sum_k X\big(j(\omega - k\omega_s)\big)$ |
| Sinc interpolation | $x_r(t) = \sum_n x(nT)\,\frac{\sin[\pi(t-nT)/T]}{\pi(t-nT)/T}$ |
| Aliased frequency | $f_{\text{alias}} = \lvert f_s - f_0\rvert$ (for $f_s/2 < f_0 < f_s$) |
| CT $\leftrightarrow$ DT frequency | $\Omega = \omega T$ |

# 10 Summary




Key takeaways:

1. **Sampling in time creates periodic copies in frequency.** The sampled spectrum is the original copy-pasted at every multiple of $\omega_s$, scaled by $1/T$.
2. **The sampling theorem** guarantees perfect reconstruction when $\omega_s > 2\omega_M$. Reconstruction uses an ideal lowpass filter (sinc interpolation in time).
3. **Aliasing** is the irreversible distortion from sampling too slowly. Higher frequencies fold back into lower ones. Use anti-aliasing filters *before* sampling.
4. **Practical reconstruction** uses zero-order holds, first-order holds, or other approximate filters.
5. **Discrete-time processing of CT signals** works because sampling preserves all information in a band-limited signal. The link: $\Omega = \omega T$.

# 11 Common Mistakes to Avoid

1. **Confusing Hz and rad/s.** $f_s > 2f_M$ (Hz) or $\omega_s > 2\omega_M$ (rad/s). Mixing units gives factor-of-$2\pi$ errors.
2. **Assuming $\omega_s = 2\omega_M$ is sufficient.** It is not (Example 22.3).

3. **Forgetting the $1/T$ scaling.** Replicas are scaled by $1/T$; the reconstruction filter compensates with gain $T$.
4. **Thinking aliasing can be undone.** It cannot. Prevent it before sampling.
5. **Claiming non-band-limited signals can't be sampled.** In practice, we band-limit with an anti-aliasing filter first.

## 12 Practice Questions

Test your understanding with these quick questions. Answers are at the bottom of the section.

1. A signal $x(t)$ has $X(j\omega) = 0$ for $|\omega| > 5000\pi$. What is the Nyquist rate? What is the maximum allowable sampling period $T$?
2. True or false: if $\omega_s = 2\omega_M$, the sampling theorem guarantees perfect reconstruction.
3. In one sentence, explain *why* sampling in time produces copies of the spectrum in the frequency domain.
4. A signal is sampled and the resulting spectrum $X_p(j\omega)$ shows overlapping replicas. Can you undo the overlap with a clever filter? Why or why not?
5. What is an anti-aliasing filter, and *where* in the signal chain does it go (before or after the sampler)?
6. Determine the Nyquist rate for $x(t) = \cos(600\pi t) + \cos(1800\pi t)$.
7. A signal with $\omega_M = 3000\pi$ is sampled with $T = 2 \times 10^{-4}\,\text{s}$. Compute $\omega_s$ and state whether aliasing occurs.
8. You sample $x(t) = \cos(2\pi \cdot 900\,t)$ at $f_s = 1000\,\text{Hz}$. What frequency does the reconstructed signal have?
9. Repeat the previous question for $f_s = 1500\,\text{Hz}$.
10. A signal has Nyquist rate $\omega_0$. What is the Nyquist rate of $x(t-5)$? What about $x(2t)$?
11. The ideal reconstruction filter has gain $T$ in its passband. Why is that factor of $T$ needed? (Hint: look at the $1/T$ scaling of the replicas.)
12. A zero-order hold is used instead of an ideal lowpass filter for reconstruction. Is the result exact? What kind of error does it introduce?
13. A movie camera shoots at 24 frames per second. A helicopter blade spins at 25 revolutions per second. Describe qualitatively what the blade looks like in the film.
14. You are told that $X(j\omega) = 0$ for $|\omega| > 10{,}000\pi$. Someone samples $x(t)$ at $f_s = 8000\,\text{Hz}$. Is the sampling theorem satisfied? If not, what is the minimum $f_s$ that works?
15. Suppose $x(t)$ is *not* band-limited (its spectrum extends to $\pm\infty$). Does this mean we can never sample it in practice? Explain what is done in real systems.

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